Autonomous Office Object Detection

My proposal for a robotic system that finds office items is a modified turtlebot4. This robot would, at a given time each night once everyone has left the office, navigate to each person's desk, extend an arm with a camera attached to it and take a photo of that person's desk. Once each desk has been photographed a program running on a nearby pc would receive the images from the turtlebot4 and then start to process each image detecting any expected items. After the detections have concluded each detection would be viewable and a full list of detected items and which desk they were found at would be made available to the users.

This can be done by either printing out the list, by emailing an admin with the list or by having the list be then passed onto a secondary software that would then hold the data until prompted or it could post the data into a work text channel such as Slack. This allows for reliable after hours tracking of items that have been misplaced or forgotten. Creating a more efficient working environment.
A basic overview of the process:

1. **Process Start:** System boot-up, checking battery levels, and verifying that all ROS 2 nodes and sensors are online.
2. **Patrol:** The robot navigates autonomously between predefined desk waypoints using SLAM.
3. **Object Manipulation:** A small 3D-printed arm extends vertically and articulates the camera into position above the desk.
4. **Perception & Recognition:** Onboard LED lights turn on to illuminate the workspace, and an RGB image is captured and fed into the detection model.
5. **Navigation:** The arm retracts, and the robot travels safely to the next desk.
6. **Report Generation:** Captured images batch-transfer over the local network to the office PC, where detections are run and the final summary list is compiled and sent out.
7. **End of Process:** The robot heads back to its charging dock and shuts down its active tasks.

Utilising the turtlebot4's LiDAR scanner we can use SLAM localisation to maneuver around the office space and to also create waypoints at each person's desk allowing us to always be in the same place for each photo. I think this would be an effective method as most office spaces are quite clear at ground level so in the event of there being an obstruction unless the office space is very small there should be another route that the turtlebot4 can take instead to reach each desk.
Making use of the network capabilities of the on board raspberry pi the transfer of photos doesn't require a person nearby to actually plug it in.
Additionally the camera would need to be accompanied with a light source ideally a small group of LEDs or another appropriately small yet effective light. Seeing as this device is meant to be deployed after hours we will have to create our own light for the camera to have visible photos.

Alternative parts in the situation that the previously stated ones are unavailable would include ultrasonic senors in place of the LiDAR scanner as 4 of them mounted in each 2D

direction would allow for a slower yet accurate manoeuvrability around the office space. In place of a raspberry pi a ESP32 could be used as it is a powerful and capable microcontroller that would work well in this given situation. An accelerometer can replace a gyrometer as long as it is sensitive enough as it will be placed atop the extendable camera arm.
In the event of the camera failing, after 5 failed detections by the software the rest of the detections will be paused as the software will run an image comparison using the “visual comparison” library on the 5 photos and if they are too similar likely due to the photos being black due to the camera not working then the rest of the process will be aborted and the final output will be a message to the administrator that the camera needs replacing.

The program would be run on a PC within the office that is connected to the network and has an appropriate GPU/CPU for this image processing. This would be a PC that is already being used in the office however in the event of the entire office only using laptops that are taken home with them a mini PC such as a NUC would suffice.

In terms of the detection protocols I will be looking for common office items such as staplers, mugs, chairs, keyboards, mice, water bottles and notebooks. These will only be marked as detected when they have a confidence score of over 85% this should exclude any false positives from occurring and impacting our overall inventory check. By using the YOLO26 large model we can get a strong level of accuracy when detecting items. Throughout the lifetime of this product it will be updated periodically with an ever growing dataset as there will be usable photos of office items that can be annotated and appended to the office items dataset. This will allow for even more accurate detections as the objects that the model is trained on will be the same objects its looking for in the office.

The safety measures for this device are somewhat lessened as this robot will only be used at the end of the day when the office is empty, meaning that the safety measures that are needed are more to do with keeping the robot safe as contact with people is unlikely. Ultrasonic sensors can be added on the robot as a backup for the collision detectors on the turtlebot4, which creates 2 failsafes for both navigation and collision as the ultrasonic sensors can be used instead of the LiDAR scanner in the event that it is no longer operational it can also be used collision sensors if the collision sensors on the turtlebot4 stop working.
The robot would also have a rather low speed limit as most desks are close together and quick acceleration and decelerations would create an unstable platform for the robot to take clear photos resulting in an increased completion time as the gyroscope module attached to the camera wouldn’t allow for the photos to be taken while it is swaying.

Bill of Materials (BOM)

Item Description	Category	Selected Model / Specification	Qty	Unit Cost (£)	Total Cost (£)

Mobile Base Platform	Chassis / Compute	TurtleBot 4 Standard (inc. RPi 4, LiDAR)	1	£1,650.00	£1,650.00
Vision Sensor	Sensor	Logitech C920 HD Webcam (1080p)	1	£45.00	£45.00
Auxiliary Sensor Array	Sensor	HC-SR04 Ultrasonic Sensors	4	£2.50	£10.00
Inertial Tracking	Sensor	MPU-6050 6-DOF IMU/Gyroscope Board	1	£5.00	£5.00
Arm Actuation	Actuator	12V High-Torque DC Gear Motor + Driver	1	£15.00	£15.00
Onboard Illumination	Lighting	5V White LED Ring Light Array	1	£8.00	£8.00
Structural Components	Chassis	3D-Printed PETG Arm Linkages & Mounts	1	£15.00	£15.00
Total Estimated Budget					**£1,748.00**